INCH-POUND

A-A-58049A
14 JANUARY 1998
SUPERSEDING
A-A-58049
20 OCTOBER 1995

COMMERCIAL ITEM DESCRIPTION

52 INCH & 100 INCH MAGNETIC PARTICLE INSPECTION UNITS

The General Services Administration has authorized the use of this commercial item description to A-A-58049 & MIL-M-6867, Type II. .

1. Scope. This commercial item description covers 52 inch and 100 inch wet horizontal magnetic particle inspection unit which shall be capable of performing conventional magnetization of ferromagnetic parts.

2. Classification.

2.1 The magnetic particle inspection units shall be of the following sizes:

Size I: 52 inch
Size II: 100 inch

3. Salient characteristics:

3.1 It shall provide alternating current (AC) and full wave rectified direct current (DC) from a three-phase AC power supply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to San Antonio ALC/TILDD, 485 Quentin Roosevelt Drive, Bldg. 171, Kelly AFB TX 78241-6425 Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6635

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3.1.1 The input power 200-240 VAC with 230 VAC nominal or 400-480 VAC with 460 VAC nominal, 50-60 Hertz, 3 phase. The unit shall have a press-to-test system indicating that the proper electricity connections have been made.

3.1.2 The rated maximum output shall be at least DC 6000 amperes with not more than 6% ripple, and for AC is 5000 amperes.

3.1.3 The unit shall be self-contained requiring only connections for electrical and compressed air supply for operation.

3.2 The unit shall be capable of performing magnetic particle inspection processes as described ASTM E1444.

3.3 The unit shall be capable of magnetizing workpiece in the longitudinal and circular directions.

3.4 The maximum contact head:

Size I 52 $\pm$ 2 inches
Size II 100 $\pm$ 2 inches

3.5 The overall length of the unit shall be as follows:

Size I no greater than 73 inches
Size II no greater than 124 inches

3.6. Instruments for inspection shall be conveniently located to permit the operator manipulation and viewing from the inspection station.

3.6.1 Digital ammeter shall display AC root mean square (RMS) amperes or DC amperes at the coil or contact. The meter shall incorporate a sample and hold the display that will be reset after magnetizing or demagnetizing current push button is depressed.

3.6.2 The current control shall be continuously variable.

3.6.3 The magnetization activation switch shall be of a type that can be activated by a knee push bar.

3.6.4 The magnetization time shall be variable from 0.20 to 2.0 seconds.

3.6.5 The headstock pressurization shall be foot switch activated

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3.6.6 The coil shall be rail mounted, have an inside diameter of 16 inches, and be capable of producing 15,000 Ampere-turns AC and DC.

3.7 The unit shall be capable of demagnetizing any workpiece previously magnetized by the same unit to less than three gausses in any directions. Both reversing DC and AC ramp demagnetization capabilities shall be built into the unit.

3.8 The operating air pressure range shall be at least 60 to 110 PSI.

3.9 The equipment shall be capable of operating at the following duty cycles indefinitely without damage to any components

Period magnetizing current "on"	Period magnetizing current "off"	Allowable percent of maximum rated magnetizing current
10 seconds	3 minutes	40 percent
5 seconds	4 minutes	100 percent

3.10 Electrical systems components, devices, wiring and insulation shall be suitable for voltages, current and duty characteristics of the circuits in which they are used shall meet the applicable requirements of the National Electrical Manufactures Association (NEMA) standards and the National Electrical Code (NEC).

3.11 Accessories. The following accessories shall be included:

3.11.1 A copper rod one inch in diameter by eighteen inches in length that will provide a low resistance path for determining magnetizing current through the contact heads.

3.11.2 Four auxiliary cables consisting of two No. 4/0 extra flexible insulated conductors with each cable being 15 feet in length, and two No. 2/0 extra flexible insulated cable conductors with each being 10 feet in length. Each conductor shall have a 4/0 connector on each end and shall have a suitable heavy duty, oil-resisting insulation.

3.11.3 A 100 Watt black light for reviewing fluorescent magnetic particle indications shall be provided

3.11.3.1 The black light shall have an intensity of at least 1000 microwatts per square centimeter at 15 inches, and shall have a peak transmission wavelength of 365 nanometers + 1.0%.

3.11.3.2 The lamp shall be mounted on swivel bracket which shall permit the lights to be directed at the parts under inspection.

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3.11.4 A hood suitable for fluorescent inspections including flame retardant curtains, a white light source and ventilation fan shall be provided.

3.11.4.1 The hood shall allow not more than 2 foot-candles (20 lux) of ambient light into the inspection area.

3.11.5 A set of replaceable copper braid pads for the headstock and tailstock contact plates.

3.11.6 The unit shall have a tank suitable for holding the suspension and particles, shall include a hose long enough to reach the entire inspection area, and a nonmagnetic low pressure spray nozzle.

3.11.7 The units shall have an agitation system capable of maintaining magnetic particles in suspension.

3.11.8 The unit shall be delivered with all equipment necessary for immediate operation (e.g. circuit breaker box, air pressure regulator, air filter).

4 Regulatory requirements. The contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. Quality assurance provisions.

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, standards, and quality assurance practices. The government reserves the right to require a government witnessed, pre-award, sample verification test at the contractor's site to insure conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be specified in the contract or order.

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7.0 Notes

7.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

Example of reference part number: A-A-58049-I or A-A-58049-II

Size I-52 Inch
Size II-100 Inch

7.2 Address for obtaining reference documents.

ASTM E 1444
American society for Testing and Materials
1916 Race Street
Philadelphia PA 19103

7.3 Ordering data.

a) Title, number, and date of this description.
b) Specify special packaging requirements.
c) Contracting officer may request proof of certification of commercial item prior to first contract delivery.

7.4 National Stock Number.

NSN	Part Number	CAGEC
6635-01-396-2798	A-A-58049 - I	81348
6635-01-361-4631	A-A-58049 - II	81348

MILITARY INTEREST:
COORDINATING ACTIVITY:

CIVIL AGENCY:
GSA-FSS

CUSTODIANS:
Air Force-99
Navy-AS
DLA-GS

PREPARING ACTIVITY:
Air Force 82

AGENT:
Air Force 99

(Project Number: 6635-0179)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-58049A	2. DOCUMENT DATE (YYMMDD) 98/01/14

3. DOCUMENT TITLE
52 INCH & 100 INCH MAGNETIC PARTICLE INSPECTION UNITS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
SA-ALC/TILDD	(1) Commercial (512) 925-6314	(2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Drive Bldg. 171, Kelly AFB TX 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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